Augmenting LLM Code Translation with Compiler Analysis for C to Triton Kernel Generation

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ABSTRACT

Emerging programming models like Triton enable developers to better exploit modern accelerators, but translating legacy code to Triton remains challenging. While Large Language Models (LLMs) show promise for code translation, they often generate incorrect or suboptimal implementations due to a lack of precise parallelization reasoning. We present TritonPilot, a compiler-assisted LLM framework for translating C loop nests to Triton kernels. TritonPilot extracts parallelization properties using compiler-based analysis and encodes them as structured analysis facts in the LLM prompt, enabling correct and performance-aware parallel code synthesis. A profiling feedback loop further improves performance by feeding GPU hardware utilization metrics back to the LLM for iterative optimization. Across PolyBench/C and TSVC kernels, TritonPilot achieves near-perfect correctness (100% and 99%), improving over LLM-only baselines. On TSVC at realistic data sizes, the generated Triton kernels achieve 1.02× median performance vs. OpenMP GPU offloading compiled by NVIDIA's production compiler on the same GPU. Profiling feedback improves 15 of 30 PolyBench kernels, with gains up to 47× on individual kernels. We further demonstrate generalization on Rodinia and ECP proxy applications. These results show that combining compiler analysis with hardware profiling feedback enables both correct and high-performance LLM-based parallel code generation.

CCS CONCEPTS

• **Software and its engineering** → **Compilers**; • **Computing methodologies** → *Parallel computing methodologies*.

KEYWORDS

LLM code generation, GPU kernels, Triton, polyhedral analysis

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1 INTRODUCTION

Accelerators such as GPUs are now essential for high-performance scientific computing, yet much existing software is still written in legacy C code that is not designed for parallel execution on modern accelerators. Translating these legacy programs to accelerator-friendly implementations is challenging, as it requires identifying parallelism, restructuring computation, and managing synchronization, tasks that require expertise in programming models and hardware.

Emerging programming models such as Triton [1] simplify GPU programming by providing higher-level abstractions for writing efficient kernels. However, mapping legacy C code to Triton still requires careful reasoning about loop-level parallelism, data dependencies, and execution order. Developers must determine which loop dimensions can be safely parallelized, how memory accesses should be organized, and when synchronization is required - decisions that are subtle and error-prone in practice.

Large Language Models (LLMs) have recently shown promise in code translation, including generating GPU kernels from high-level descriptions [2, 3]. However, LLMs lack precise reasoning about program semantics, especially data dependencies. As a result, they often produce incorrect or inefficient code, including unsafe parallelization that leads to data races, suboptimal kernel structures, and missing synchronization.

In this paper, we present TritonPilot, a system that augments LLM-based translation with compiler analysis. TritonPilot extracts parallelization properties from C loop nests using polyhedral analysis (PET) [4] and LLVM dependence analysis [5], and encodes them as structured guidance in the LLM prompt. This allows the model to generate correct and performance-aware Triton code without requiring it to perform dependence analysis itself.

We evaluate TritonPilot on a diverse set of benchmarks and application kernels on an NVIDIA RTX 3090 GPU. Across PolyBench/C and TSVC loop kernels, TritonPilot achieves near-perfect correctness (100% and 99%), significantly improving over LLM-only baselines. At realistic data sizes, the generated Triton kernels match the performance of OpenMP GPU offloading compiled by NVIDIA's `nvc` compiler (1.02× median on the same GPU). We further demonstrate generalization on kernels from Rodinia and ECP proxy applications. This paper makes the following contributions:

- A compiler-assisted LLM framework for translating legacy C loop nests to Triton, with a unified analysis module that encodes parallelization constraints as structured facts rather than pattern-specific templates.

- A profiling-guided optimization loop that feeds GPU hardware utilization metrics (from NVIDIA Nsight Compute) back to the LLM, enabling iterative performance improvement while preserving correctness.
- Empirical results showing that compiler analysis ensures correctness (100% on PolyBench/C) and that profiling feedback recovers the performance of hand-tuned prompt templates while maintaining generality.

2 OUR APPROACH

Figure 1 shows the TritonPilot workflow. Given a C source file, TritonPilot extracts loop nests, analyzes their parallelization properties, constructs a structured prompt, invokes an LLM to generate Triton code, and optionally refines performance using GPU profiling feedback. TritonPilot is implemented in approximately 14K lines of Python, using PET/isl [8] for polyhedral analysis and LLVM 17 [5] for fallback dependence analysis.

2.1 Unified Analysis

TritonPilot calls PET [4] once per kernel to extract a polyhedral representation (iteration domains, access relations, schedules as ISL objects). From this single invocation, it derives all parallelization constraints:

(i) Parallelization safety. For each loop dimension, TritonPilot checks for cross-iteration dependences by composing ISL access relations: given read map R and write map W to the same array, the conflict relation $R \circ W^{-1}$ is intersected with the iteration domain and the identity map is subtracted. A non-empty result marks the dimension as unsafe. *(ii) WAR dependences.* Write-after-read conflicts are identified when an array is both read and written at different iteration points, requiring array cloning or sequential execution. *(iii) Memory access patterns.* Stencil patterns (neighboring-element reads), triangular access (`j < i`), and cross-phase dependencies (multiple phases writing different arrays). *(iv) Reductions.* Scalar writes combined with array reads indicate reductions, mapped to Triton primitives (`tl.sum()`, `tl.dot()`).

When polyhedral analysis fails (e.g., non-affine accesses), TritonPilot falls back to LLVM's `DependenceAnalysis`, parsing direction vectors to determine which loop levels carry dependences.

2.2 Prompt Construction

All analysis results are stored in a unified JSON schema with fixed fields regardless of what patterns the kernel exhibits. A single, pattern-agnostic function renders the analysis into structured text. This design principle—*tell the LLM what is true about the kernel, rather than prescribing what to do*—improves scalability: adding a new analysis pass requires only extending the JSON schema, not modifying prompt templates. The prompt includes the C source, analysis facts (e.g., "`j` is safe to parallelize; `t` has cross-phase dependencies; stencil pattern detected"), the required function signature, and Triton coding rules.

2.3 Generation and Iterative Refinement

TritonPilot invokes the LLM (Claude Sonnet 4) and validates the generated kernel against the C reference using identical inputs and FP32 tolerance checks. On failure, errors are classified into four categories: compilation/runtime errors, numerical errors, missing synchronization barriers, and low performance ($<0.1\times$). Each triggers a targeted retry prompt. This repeats up to 10 attempts, with a context reset at attempt 6 to mitigate error accumulation.

2.4 Profiling-Guided Optimization

After correctness validation, TritonPilot optionally profiles the kernel with NVIDIA Nsight Compute (NCU), collecting SM throughput, memory throughput, occupancy, and L1/L2 cache hit rates. From these metrics, it classifies the bottleneck as *compute-bound* (high SM utilization), *memory-bound* (high memory throughput), or *latency-bound* (both low, indicating insufficient parallelism).

The bottleneck classification, raw metrics, and current code are fed back to the LLM with bottleneck-specific optimization guidance: memory-bound kernels are guided toward tiling and coalescing; latency-bound kernels toward increased grid size. The optimized kernel is re-validated for correctness and kept only if faster; otherwise, the previous version is retained. This loop runs up to 3 iterations, with NCU results cached when the implementation is unchanged.

3 EVALUATION SETUP

Platforms. All experiments are conducted on a GPU workstation equipped with an NVIDIA RTX 3090 GPU (24 GB memory, 936 GB/s memory bandwidth) and a dual-socket Intel Xeon Gold 5218R CPU (80 logical cores). We use PyTorch 2.5.1, Triton 3.1.0, CUDA 12.4, GCC 11.4.0, and Claude Sonnet 4 [7] as the underlying LLM.

Benchmarks. We evaluate on three benchmark suites: (1) *PolyBench/C* [6]: 30 numerical kernels (stencils, linear algebra, data mining) at default sizes (N=60–120) and 8× scale (N=480–960); (2) *TSVC* [15]: 151 loop kernels covering linear traversals, reductions, indirect addressing, and loop-carried dependences, benchmarked at 32 MB and 3200 MB per array; (3) 8 application kernels from Rodinia [13] (SRAD, lud, gaussian, lavaMD, hotspot, pathfinder) and ECP proxy applications [14] (miniMD LJ force, miniFE SpMV).

Baselines. The primary comparison is against *OMP GPU*—OpenMP GPU offloading compiled by NVIDIA `nvc` with managed memory, running on the *same* GPU as Triton. This isolates the quality of LLM-generated Triton code from hardware differences. We additionally report against *OMP CPU* (80-thread OpenMP) and *ST* (single-threaded GCC `-O3`).

Evaluation methodology. All speedups measure kernel-only execution time with data pre-staged on the GPU, averaged over 10–50 iterations after warmup. For the correctness ablation, we compare *with analysis* (WA) vs. *no analysis* (NA, LLM receives only the C source). For profiling feedback, we

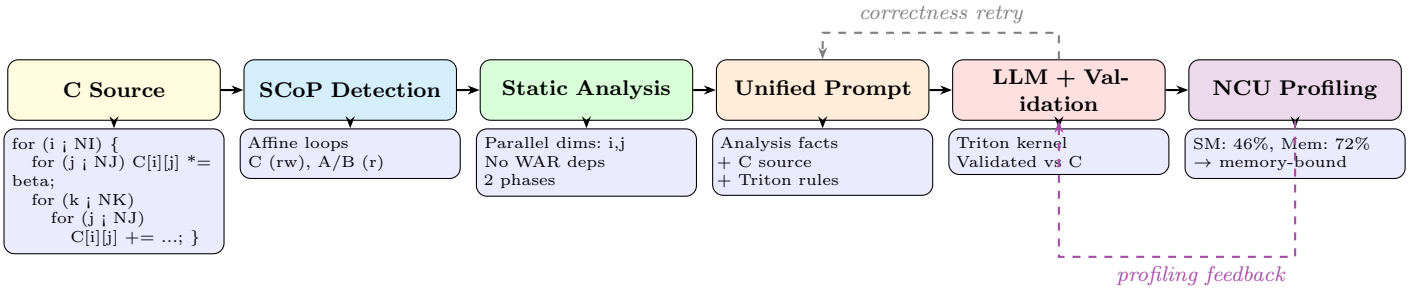

Figure 1: The TritonPilot workflow. Dashed arrows show iterative loops: the gray loop retries on correctness failures; the violet loop feeds NCU profiling metrics back to the LLM for performance optimization.

Table 1: PolyBench/C results. WA = with analysis, NA = no analysis.

Metric	WA	NA
Pass rate	30/30 (100%)	28/30 (93%)
Median speedup	2.07×	1.06×
Mean speedup	2.58×	1.52×
Kernels > 1×	20/28	14/28
Head-to-head (28 common):		
WA wins (>10% faster)	12	
NA wins (>10% faster)	6 (3 race conditions)	
Ties (within 10%)	10	

Table 2: Representative profiling feedback improvements (PolyBench 1×). "Before" = first correct kernel; "After" = best after NCU-guided iterations.

Kernel	Before	After	Bottleneck
covariance	0.15×	5.40×	latency
deriche	0.11×	5.14×	latency
3mm	0.12×	4.24×	latency
correlation	0.29×	3.38×	latency
adi	0.10×	2.07×	latency
gemver	0.16×	1.94×	latency
gemm	2.41×	3.10×	latency
fdtd_2d	0.85×	1.80×	latency
durbin	0.30×	1.47×	latency

report the initial speedup (first correct kernel) and the final speedup after up to 3 NCU-guided optimization iterations.

4 EXPERIMENTAL RESULTS

4.1 PolyBench/C (30 Kernels)

Table 1 summarizes results at standard (1×) problem sizes. All 30 kernels pass with analysis guidance; without it, two stencil kernels fail from parallelizing dimensions with carried dependences. Compute-bound kernels show strong speedups (`heat_3d`: 10.27×, `doitgen`: 7.40×, `covariance`: 5.40×, `deriche`: 5.14×).

In the head-to-head comparison, 3 of 6 apparent NA wins are *false wins*: the unguided LLM parallelizes dimensions with carried dependences, producing code that runs faster but computes incorrect results. For example, in `lu`, the NA code launches one GPU thread per row (`grid=(N,)`), but row i depends on row k ($k < i$), causing output differences up to 10^{33}.

Scaling. At 8× sizes (N: 60→480, matrices: 120→960), median speedup rises to 30.73× over single-threaded C and 8.57× over 80-thread OpenMP, confirming genuine advantage beyond CPU parallelism.

4.2 Profiling-Guided Optimization

Profiling feedback improved 20 of 28 passing kernels (Table 2 shows representative gains). The largest improvements occur on latency-bound kernels where the initial implementation has insufficient parallelism: the NCU metrics reveal low SM and memory utilization (<20%), and the LLM restructures the kernel to use more parallel resources. For example, `covariance` improves from 0.15× to 5.40× (36× gain) as the LLM transforms a single-CTA sequential implementation into a multi-CTA parallel grid after receiving the latency-bound diagnosis.

Correctness is preserved across all profiling iterations: each optimized kernel is re-validated, and attempts that break correctness are automatically rejected. The majority of optimization attempts do break correctness—the LLM frequently introduces race conditions when restructuring parallelism. This confirms that single-shot prompting is insufficient for reliable optimization and motivates integration with search-based or RL-based methods that can explore many candidates.

4.3 TSVC (151 Kernels)

TritonPilot passes 150/151 TSVC kernels (99.3%), 120 on first attempt. At 32 MB per array, LLM-generated Triton achieves 67× over single-threaded C, 10.7× over 80-thread OpenMP CPU, and **1.02× vs. OpenMP GPU offloading** (`nvc -mp=gpu`) on the same GPU—matching production compiler quality.

Table 3: Application kernels: Triton vs. OMP GPU offloading (`nvc`) on the same GPU.

Source	Kernel	Tri/OMP GPU	WA/ST	NA/ST
Rodinia	SRAD	24.9×	24.9×	3.5×
Rodinia	gaussian	249×	3.4×	3.6×
miniMD	LJ force	6.8×	25.8×	23.0×
miniFE	SpMV	3.1×	9.0×	7.3×
Rodinia	lud	—[a]	13.2×	15.5×
Rodinia	lavaMD	—[a]	1.5×	0.6×

[a] OMP GPU offloading (`nvc`) fails to compile these kernels.

4.4 Application Kernels

To evaluate generalization, we apply TritonPilot to 6 Rodinia and 2 ECP kernels (Table 3). All 8 pass with analysis guidance. On kernels with OMP GPU baselines, Triton is 3–249× faster than `nvc`'s offloading, due to Triton's block-level model avoiding per-timestep launch overhead (gaussian: 511 timesteps × 3 launches = 1533 offloads). Analysis guidance is most impactful on multi-phase kernels: SRAD improves from 3.5× (NA) to 24.9× (WA) via phase decomposition.

5 RELATED WORK

LLM Code Generation for GPUs. Chen et al. [2] find frequent correctness issues in ChatGPT-generated CUDA kernels. KernelBench [3] reports 20–40% functional correctness for LLM-generated GPU kernels without guidance. Recent works improve generation through RL fine-tuning: Dr. Kernel [16] uses profiling-based rewards, Tehrani et al. [17] RL-fine-tune GPT-5, and DICE [18] applies RL to diffusion LLMs. Search-based methods keep the LLM frozen: AVO [19] uses LLM agents as evolutionary operators, K-Search [20] decouples planning from code instantiation, and OptiML [21] uses MCTS with profiler feedback. KernelBlaster [22] enables cross-task learning via persistent memory. For non-GPU targets, AscendKernelGen [23] and AscendCraft [24] address NPU kernels. These works focus on *how* to generate/optimize kernels; our work is orthogonal, addressing *what* parallelism is safe via compiler analysis. The two directions are complementary: compiler-derived constraints could serve as correctness rewards in RL frameworks or safety priors in search.

Polyhedral Compilation and Auto-Scheduling. PPCG [12] generates CUDA from affine loop nests but produces low-level code that lacks Triton's block-level optimizations. TVM [9] and Halide [10] use auto-scheduling with learned cost models. TritonPilot is complementary: it uses polyhedral analysis to extract *what* parallelism is safe, then delegates the *how* to the LLM, which can handle non-affine constructs, irregular access patterns, and Triton abstractions that polyhedral compilers cannot target directly.

6 CONCLUSION

We presented TritonPilot, a system that combines compiler-based analysis with LLM code generation for translating C loop nests to Triton GPU kernels. A unified analysis module calls PET once per kernel and derives all parallelization constraints (safe dimensions, WAR dependences, reductions, memory patterns) from the single polyhedral representation. These are rendered as structured facts in the LLM prompt via a pattern-agnostic renderer, replacing pattern-specific templates.

TritonPilot achieves 100% correctness on PolyBench/C (30/30) and 99% on TSVC (150/151). The generated Triton code matches production compiler quality: 1.02× median vs. `nvc`'s OpenMP GPU offloading on the same GPU. It generalizes to 8 application kernels from Rodinia and ECP proxy apps (up to 249× vs. OMP GPU).

A profiling feedback loop using NCU hardware metrics improved 20 of 28 PolyBench kernels, with gains up to 36×. However, the LLM frequently breaks correctness during optimization, confirming that single-shot prompting is insufficient. Future work will integrate compiler analysis as a correctness oracle and NCU metrics as a performance reward within search-based or RL-based optimization frameworks.

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